

练习十九 静电场 (一)

1. 2 a

2. $\frac{\rho r}{3\varepsilon_0}$; $\frac{\rho a^3}{3\varepsilon_0 r^2}$

3. (3); 4. (1)

5. $E = \int_d^{d+l} \frac{\lambda dx}{4\pi\varepsilon_0 x^2} = 675(N/C)$

6. $E = \frac{\lambda}{2\pi\varepsilon_0 r}$ $dF = Edq = \frac{\lambda}{2\pi\varepsilon_0 r} \cdot \lambda dr$

$$F = \int_R^{R+L} \frac{\lambda^2 dr}{2\pi\varepsilon_0 r} = \frac{\lambda^2}{2\pi\varepsilon_0} \ln \frac{R+L}{R}$$